

**MEDIGUARD: AN ARTIFICIAL INTELLIGENCE-DRIVEN
PATIENT HEALTH RECORD
MANAGEMENT, PRESCRIPTION ANALYSIS, CONSENT
GOVERNANCE,
AND CLINICAL SAFETY MONITORING SYSTEM**

ABSTRACT

A patient-centric system for secure medical record management, artificial intelligence-assisted prescription digitisation, consent-governed clinician access, clinical safety monitoring, and tamper-evident audit logging is disclosed. A patient authenticates through a Vault Application (119) on a Patient Mobile Terminal (118) and uploads handwritten or printed prescriptions to a MediGuard Artificial Intelligence Processing Center (101). A Pre-Processing Unit (104) normalises image contrast prior to text extraction by a Vision Optical Character Recognition Engine (106), whose output is structured into a clinical schema by a Data Structuring Module (107). A Medicine Information Module (109) derives bilingual patient instructions in English and Tamil for each identified medication. The patient governs clinician access through an explicit consent framework; Authorized Clinicians (117) may additionally invoke a time-limited emergency break-glass override via the Secure Network (116). A clinical safety module cross-references the patient's active medications and allergy profile to surface drug-drug and drug-allergy conflict alerts. All records are sealed by an Advanced Encryption Standard 256 Encryption Unit (110) before storage in Firebase Cloud Storage (112) and Firestore Database (113). Every access event, consent change, and override is logged by an Audit Log Generator (111) within Consent and Audit Logs (114), forming a chronologically ordered, patient-accessible trust record.

Reference is made to the accompanying **FIG. 1**, which illustrates the complete architecture of the MediGuard Artificial Intelligence Processing Center (101) and all subsystems described herein.

